COURSE TITLE : INSTRUMENT TRANSDUCERS
COURSE CODE : 3214
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER: 60
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Basics of Transducers and Resistance transducer	15
2	Inductance transducers and Magnetic transducers	15
3	Capacitance, Piezo elecric and Photo electric transducers	15
4	Radiation Detectors and Special types of Transducers	15
TOTAL		**60**

COURSE OUTCOME

Module	G.O.	On completion of the study of this module the student will be able
1	1	To understand the basics of transducers
	2	To comprehend the working of Resistance transducers
2	3	To understand the working of different types of Inductance transducers
	4	To understand the working of different types of Magnetic transducers
3	5	To understand the working of Capacitance transducers
	6	To comprehend the working of Piezo electric transducers
	7	To understand the principle of operation of Photo electric transducers
4	8	To comprehend the working of Radiation detectors
	9	To know the Special types of transducers used in instrumentation applications

On completion of the study the student will be able

MODULE I **BASICS OF TRANSDUCERS AND RESISTANCE TRANSDUCER**

1.1.0 To understand the basics of transducers

1.1.1 To define transducer
1.1.2 To classify different types of transducers like active , passive , analog and digital .
1.1.3 To differentiate between sensors and transducers
1.1.4 To compare mechanical and electrical transducer with examples

1.2 .0 To comprehend the working of Resistance transducers

- 1.2.1 To describe the principle of resistive transducers
- 1.2.2 To explain the principle of operation of linear and rotary potentiometer
- 1.2.3 To derive the sensitivity of linear and rotary potentiometer
- 1.2.4 To describe loading effect
- 1.2.5 To explain the Principle of operation of strain gauges
- 1.2.6 To derive the expression for gauge factor
- 1.2.7 To list the types of strain gauges
- 1.2.8 To state the advantages and disadvantages of semiconductor strain gauges
- 1.2.9 To explain strain gauge bridge circuit

<u>MODULE II</u> INDUCTANCE TRANSDUCERS AND MAGNETIC TRANSDUCERS

2.1.0 To know the working of different types of Inductance transducers

- 2.1.1 To explain the principle of variable inductance transducer
- 2.1.2 To classify Inductive transducers
- 2.1.3 To explain variable reluctance and variable eddy current inductive transducer
- 2.1.4 To describe the construction and working of LVDT
- 2.1.5 To describe pressure and weight measurement using LVDT.

2.2.0 To know the working of different types of Magnetic transducers

- 2.2.1 To list the types of Magnetic transducers.
- 2.2.2 To explain the working of search coils.
- 2.2.3 To explain the operation of magneto resistive transducer.
- 2.2.4 To distinguish between magneto resistive and magnetostictive transduce
- 2.2.5 To define hall effect
- 2.2.6 To explain principle of operation of hall effect transducer

<u>MODULE III</u> CAPACITANCE, PIEZO ELECRIC AND PHOTO ELECTRIC TRANSDUCERS

3.1.0 To understand the Capacitance transducers

- 3.1.1 To describe the principle of variable capacitive transducer
- 3.1.2 To explain the working of variable area, variable distance, and variable dielectric type capacitive transducers
- 3.1.3 To illustrate differential capacitive transducer
- 3.1.4 To describe the applications of variable capacitive transducer

3.2.0 To comprehend Piezo electric transducers

- 3.2.1 To define piezo electric effect
- 3.2.2 To explain principle of operation of piezo electric transducers
- 3.2.3 To describe the equivalent circuit
- 3.2.4 To list the applications – pressure measurement and accelerometer

3.3.0 To know Photo electric transducers

- 3.3.1 To define photoelectric effect
- 3.3.2 To Illustrate the working principle of photo emissive cell
- 3.3.3 To describe the working principle of Conductive cell
- 3.3.4 To explain the working principle of photo voltaic cell

3.3.5 To describe the working principle of photo multipier tube
3.3.6 To describe the principle of operation of solar cells.
3.3.7 To list the applications of photo electric transducers

MODULE IV RADIATION DETECTORS AND SPECIAL TYPES OF TRANSDUCERS

4.1.0 To comprehend Radiation detectors

4.1.1 To explain the construction and operation of ionization chamber
4.1.2 To describe the construction and operation of propotional counter
4.1.3 To explain the construction and operation of gieger muller counter
4.1.4 To Illustrate the construction and operation of scintillation counter
4.1.5 To describe the working of solid state transducers for radiation
4.1.6 List the applications of radiation detectors.

4.2.0 To know the working of Special types of transducers used in instrumentation applications

4.2.1 To explain the working principle of Ultrasonic transducers
4.2.2 To describe the working principle of inductive type Proximity sensors
4.2.3 To explain the working of capacitive type proximity sensors
4.2.4 To Illustrate the working principle of Smart sensors and smart transmitters
4.2.5 To explain the functions of IC sensors AD 590 and LM 35
4.2.6 To brief about nano sensors
4.2.7 To brief about MEMS

CONTENT DETAILS

MODULE I

Transducer – classificatio- active, passive, analog and digital - sensors and transducers - mechanical and electrical transducer with examples **-** principle of resistive transducers - principle of operation of linear and rotary potentiometer- sensitivity of linear and rotary potentiometer - loading effect - strain gauges - expression for gauge factor - types of strain gauges - advantages and disadvantages of semiconductor strain gauges - strain gauge bridge circuit.

MODULE II

Variable inductance transducer - Classification of Inductive transducers - variable reluctance and variable eddy current inductive transducer - construction and working of LVDT - applications of LVDT - Different types of Magnetic transducers - Working of search coils - Operation of magneto resistive transducer - Magneto resistive and magnetostictive transducer - Hall effect transducer.

MODULE III

Variable capacitive transducer - Working of variable area, variable distance, and variable dielectric type capacitive transducers - Differential capacitive transducer - Applications of variable capacitive transducer - Piezo electric effect - principle of operation of piezo electric transducers - equivalent circuit- pressure measurement and accelerometer applications- Photoelectric effect - working principle of photo emissive cell , Conductive cell , photo voltaic cell , photo multipier tube , solar cells - applications of photo electric transducers .

MODULE IV

Construction and operation of ionization chamber – Propotional counter - Gieger muller counter - Scintillation counter - Working of solid state transducers for radiation - applications of radiation detectors.-Working principles of Ultrasonic transducers, Inductive type Proximity sensors, Capacitive type proximity sensors , Smart sensors and smart transmitters. IC sensors AD 590 and LM 35- Nano sensors- MEMS. asics of Tranducers and Resistance transducesics of transdScope ofInstrumentationgineering

REFERENCES

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2. D.V.S.Murthy , Transducers and Instrumentation , PHI Pvt Ltd
3. J.B.Gupta , Electrical and Electronic Measurements and instrumentation
4. Ranganathan.S , transducer engineering , Allied publishers
5. akra Chowdhari , Instrumental methods of Analysis